Computer Networks

Physical Layer

Unit 3

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- Signals
- Transmission Modes
- Data Transmission (Method)

Functions of Physical Layer

- **Bit-by-Bit Transmission**
- **Transmission Modes**
- **Encoding and Decoding**
- **Data Control**
- **Signal Transmission**
- **Modulation and Demodulation**

Signals

- **Signals:** The physical manifestation of data (e.g., voltage levels for digital, continuous waves for analog).
    - **Analog:** Continuous waves (e.g., sound, radio).
    - **Digital:** Discrete pulses (0s and 1s).

Signals

Similar to data, a signal can be analog or digital. An analog signal can have infinite number of values in a given range, on the other hand a digital signal has limited number of values in a given range. The following digram shows analog and digital signals.

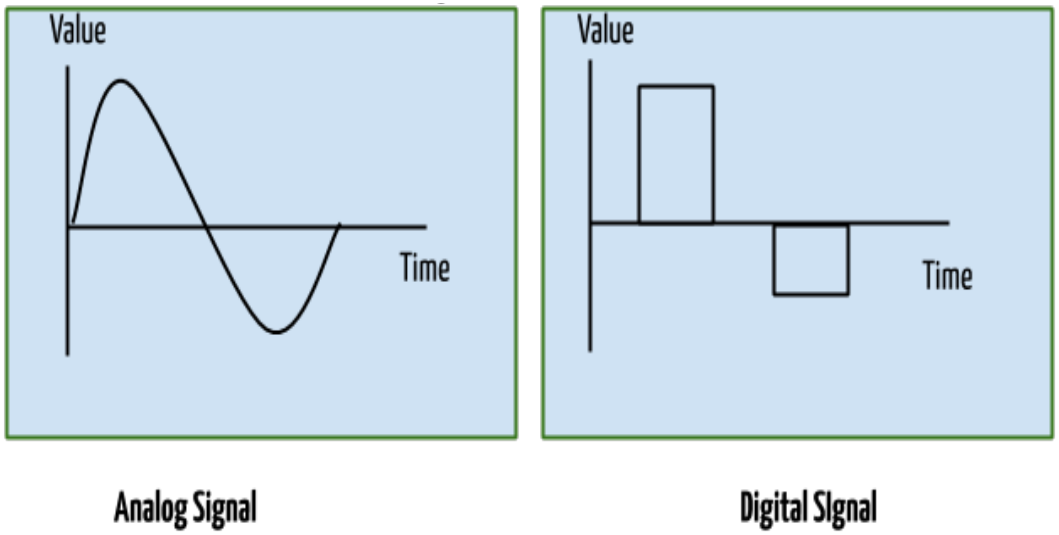
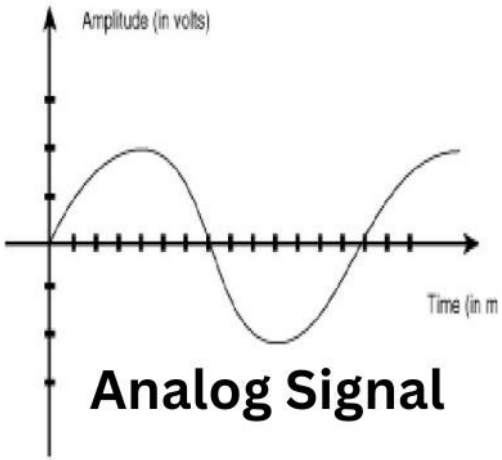
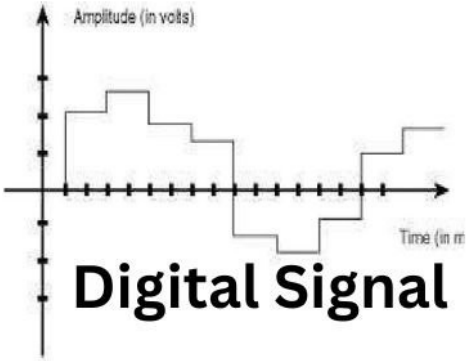


Figure 1. Signals

Signals





Header label	Analog Signal	Digital Signal
Signal	1) In an Analog, signals are continuous.	1) In a Digital, Signals are discrete.
Transformation	2) In analog systems electronic circuits are used for the transformation of signals.	2) In Digital Signals, the transformation is done using the logic circuit.
Transmission	3) Data transmission is not of high quality.	3) Data transmission has high quality.
Flexibility	4) In an Analog signal, their hardware is not flexible.	4) In Digital signals, their hardware is not flexible.
Noise	5) Analog signals are more likely to get affected and result in reduced accuracy.	5) Digital signals are discrete time signals that are generated by digital modulation.
Power Consumptions	6) Analog signals use more power.	6) Digital signals use less power compared to analog.
Waves	7) It is denoted by the sine waves.	7) It is denoted by the square form.
Example	8) Human Voice, Tape recorder, Temperature, etc.	8) Mp3 players, Digital phones, computers, etc.

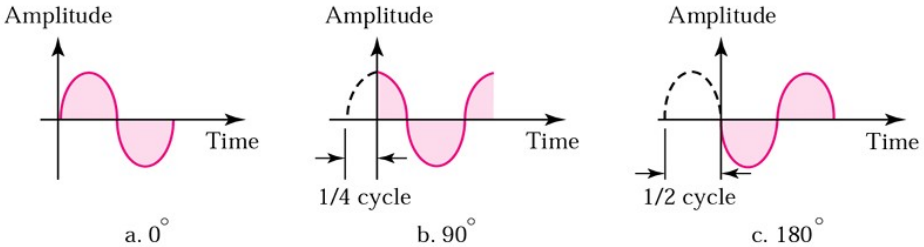
Signals

- Peak amplitude of a signal is the absolute value of its highest intensity, proportional to its energy it carries. It is measured in volts.
- Period refers to the amount of time in seconds, a signal needs to complete 1 cycle.
- Frequency is the rate of change with respect to time. Change in a short span of time means high frequency. Change over a long span of time means low frequency.

  Frequency : refers to the number of periods in 1s.

  $$f=1/T \text{ and } T=1/f$$

  Frequency and period are the inverse of each other
- Phase describes the position of the waveform relative to time 0. It indicates the states of the first cycle Three sine waves with same amplitude and frequency But diff phases.



Transmission Modes (Directionality)

- **Simplex:** One way only (e.g., TV broadcast).
- **Half-Duplex:** Both ways, but one at a time (e.g., walkie-talkie).
- **Full-Duplex:** Both ways simultaneously (e.g., phone call).

Data Transmission (Method)

- **Serial Transmission:**
  - **How:** One bit after another over a single wire.
  - **Pros:** Reliable, cheap, good for long distances (e.g., USB, Ethernet).
  - **Cons:** Slower.
  - **Types:**
    - **Asynchronous:** Byte-by-byte with start/stop bits; simpler, less overhead (e.g., older modems).
    - **Synchronous:** Blocks of data with sync patterns; faster, accurate clock needed (e.g., high-speed networking).
- **Parallel Transmission:**
  - **How:** Multiple bits (e.g., 8 bits/byte) simultaneously via multiple wires.
  - **Pros:** Faster (e.g., older printers, internal bus).
  - **Cons:** Less reliable, more complex wiring, shorter distances due to signal interference.

Data Transmission (Method)

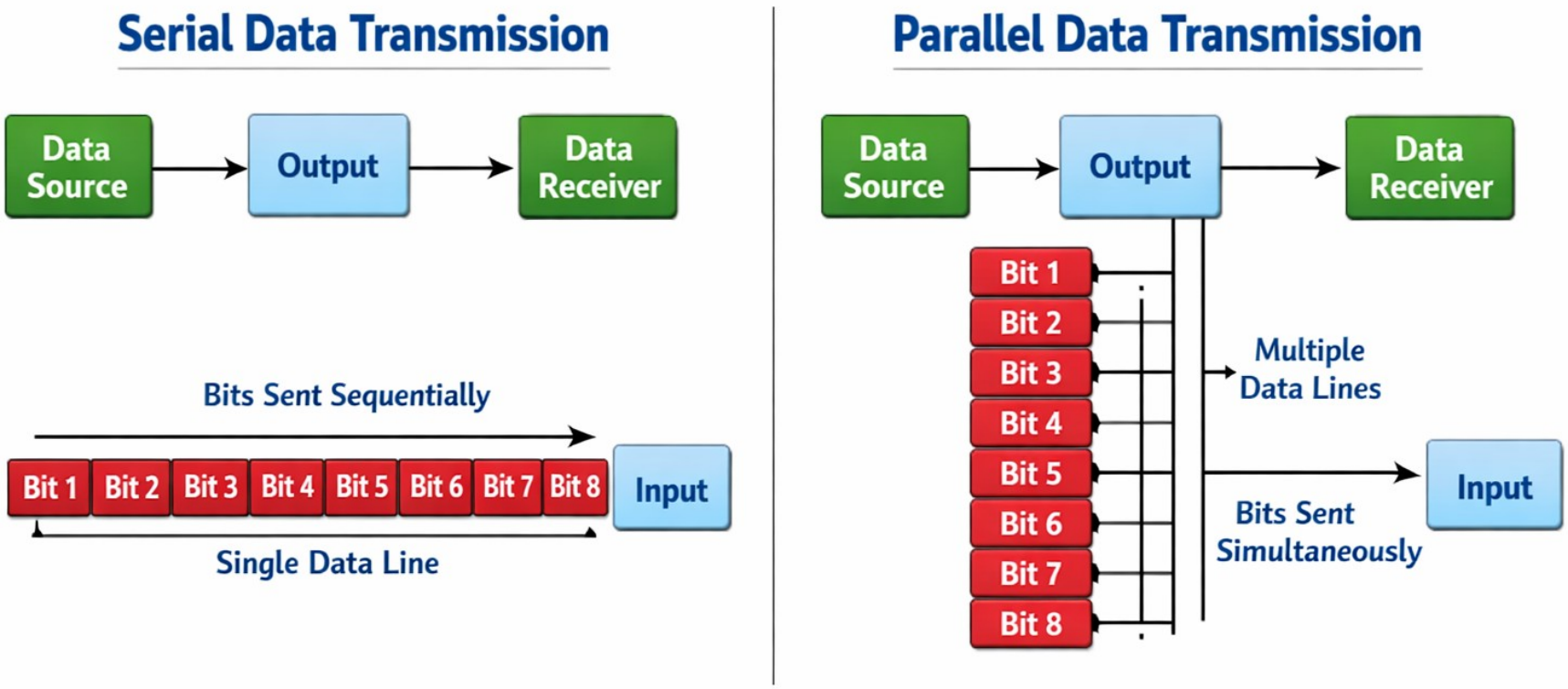


Figure 2. Data Transmission

Digital Data Transmission

Line Coding: Converting digital bits (1s and 0s) into a digital signal (voltage levels) for transmission.

- **Techniques :** Unipolar (uses one voltage level), Polar (uses two levels), Bipolar (uses three levels, like AMI).

Properties of Line Coding

- As the coding is done to make more bits transmit on a single signal, the bandwidth used is much reduced.
- For a given bandwidth, the power is efficiently used.
- The probability of error is much reduced.
- Error detection is done and the bipolar too has a correction capability.
- Power density is much favorable.
- The timing content is adequate.
- Long strings of **1s** and **0s** is avoided to maintain transparency.

Line Coding

Main Types of Line Coding

- **Unipolar:** Uses a single voltage level (e.g., positive voltage for '1', zero for '0'). Simple but inefficient and prone to DC issues, so rarely used.
- **Polar:** Uses two voltage levels (e.g., positive for '1', negative for '0'). More efficient than unipolar.
- **Bipolar:** Uses three voltage levels (positive, negative, zero). Offers advantages like no DC component and better error detection.

Line Coding

Common Techniques

- **NRZ (Non-Return-to-Zero):** Signal stays at the same level for the entire bit duration (e.g., NRZ-L uses different levels for '0'/'1', NRZ-I inverts for '1').
- **RZ (Return-to-Zero):** Signal returns to zero voltage in the middle of the bit time, aiding synchronization but uses more power.
- **Biphase (Manchester & Differential Manchester):**
  - **Manchester:** A transition (high-to-low or low-to-high) occurs in the middle of each bit, providing excellent self-synchronization and no DC component.
  - **Differential Manchester:** Transition occurs at the beginning of the bit, with an inversion for '1' and no inversion for '0' (or vice versa), similar to NRZ-I but with mid-bit transitions.

Line Coding

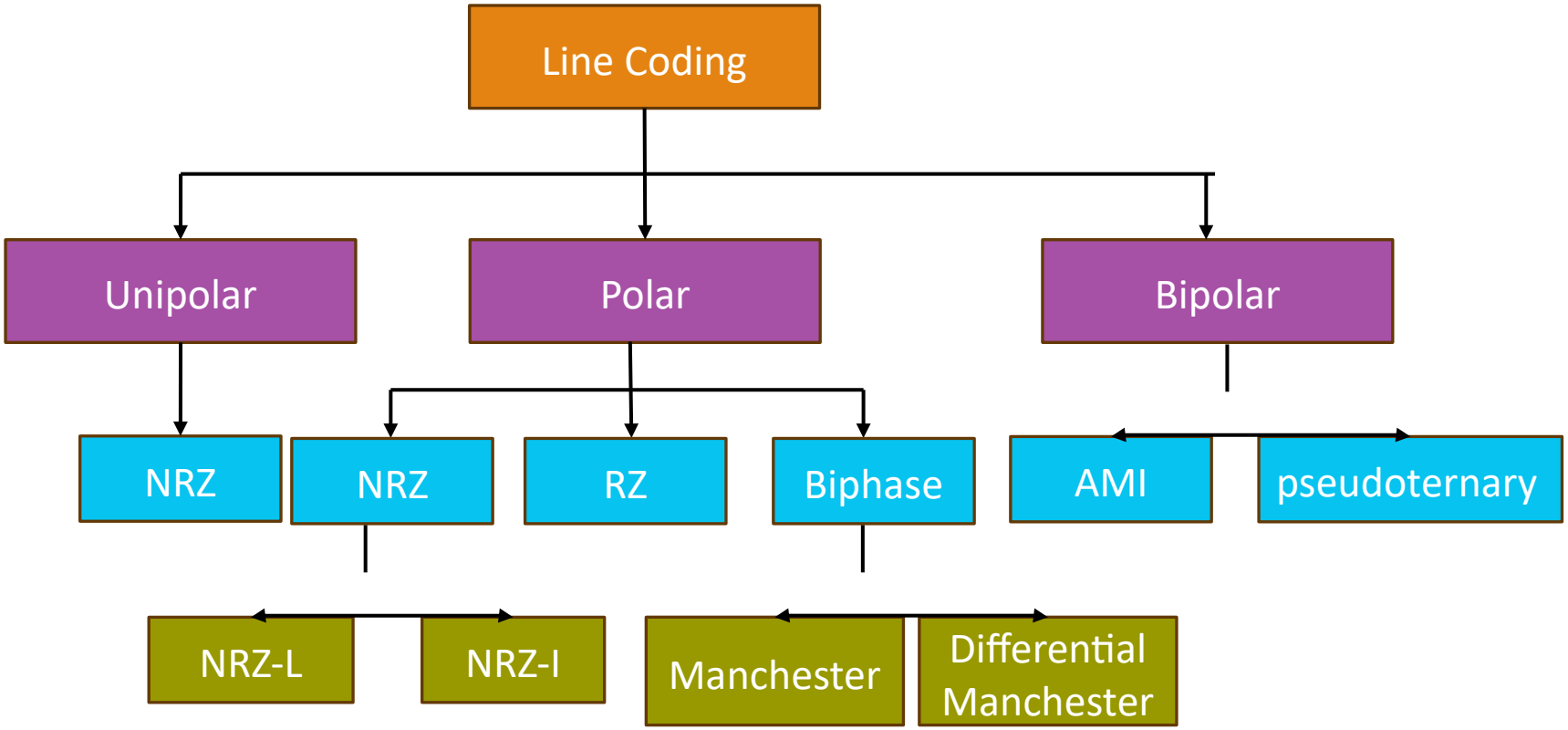


Figure 3. Classification of Line Coding Scheme

Line Coding

Line Coding Scheme	Voltage Levels	Advantages	Disadvantages
Unipolar	One voltage level positive	Simple to implement low complexity	Poor synchronization long zero sequences
Polar	Two voltage levels positive and negative	More efficient than unipolar	Synchronization issues in long zeros
Bipolar	Three voltage levels positive negative zero	No DC component improved error detection	More complex to implement

Digital Data Transmission

Unipolar Signaling

Unipolar signaling uses only one voltage level to represent the binary data. Typically, binary 1 is represented by a positive voltage or sometimes zero and binary 0 is represented by zero voltage.

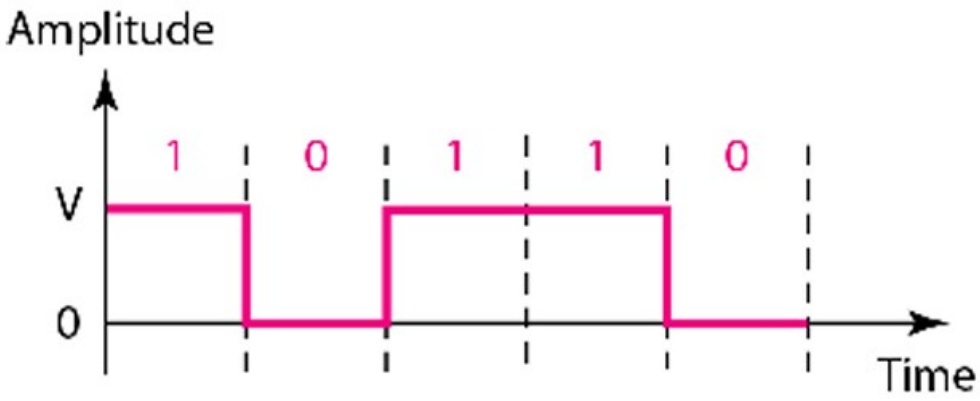
Unipolar Non Return to Zero (NRZ)

In unipolar NRZ coding a binary 1 is represented by a positive voltage and binary 0 is represented by zero voltage. The signal does not return to zero after each bit which makes it simpler to implement but can cause long sequences of zero voltages that lead to synchronization issues at the receiver.

- **Advantages**
  - Simple and easy to implement
  - Low complexity
- **Disadvantages**
  - Lack of synchronization due to long zeros
  - Poor error detection

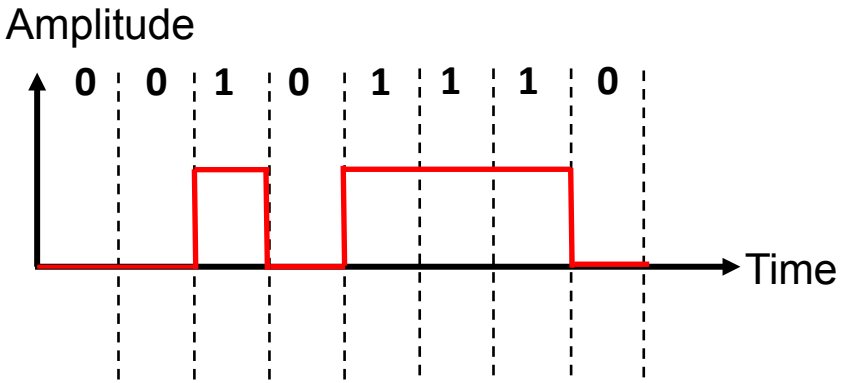
Unipolar Non Return to Zero (NRZ)

Data: 10110



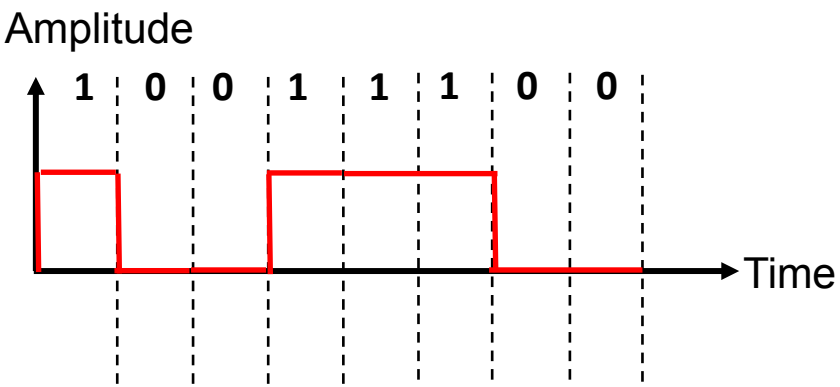
Unipolar Non Return to Zero (NRZ)

Data: 00101110



Unipolar Non Return to Zero (NRZ)

Data: 10011100



Digital Data Transmission

Polar Signaling
Polar signaling involves two voltage levels, one positive and one negative to represent binary data. Binary 1 is typically represented by a positive voltage and binary 0 is represented by a negative voltage. Polar signaling is used to improve signal integrity and synchronization.

- **Polar NRZ**
- **Polar RZ**
- **Polar Biphase**

Polar NRZ

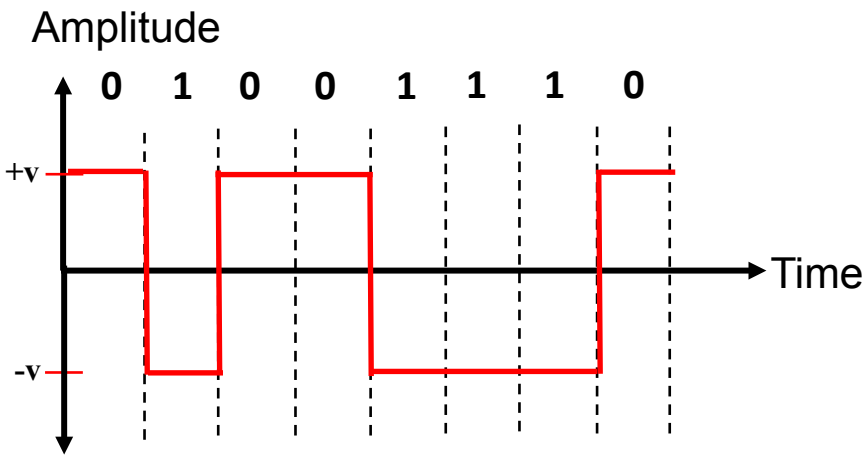
The voltages are on both sides of the time axis. Polar NRZ scheme can be implemented with two voltages. E.g. +V for 1 and -V for 0. two voltages. E.g. +V for 1 and -V for 0. There are two versions:

NZR - Level (NRZ-L) - positive voltage for one symbol and negative for the other

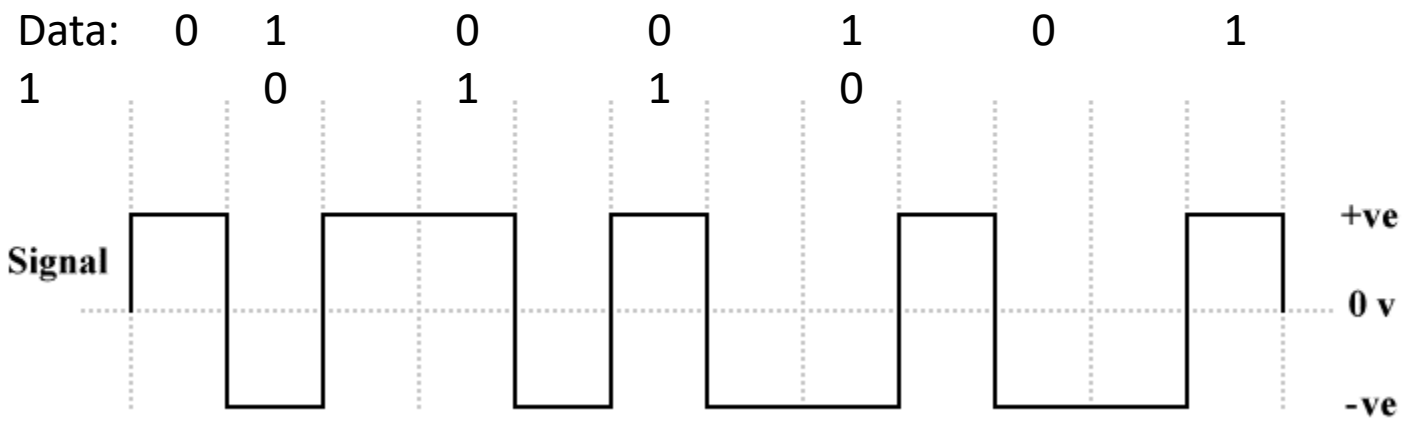
NRZ - Inversion (NRZ-I) - the change or lack of change in polarity determines the value of a symbol. E.g. a “1” symbol inverts the polarity a “0” does not.

Polar NRZ - Level (NRZ-L)

Data: 01001110

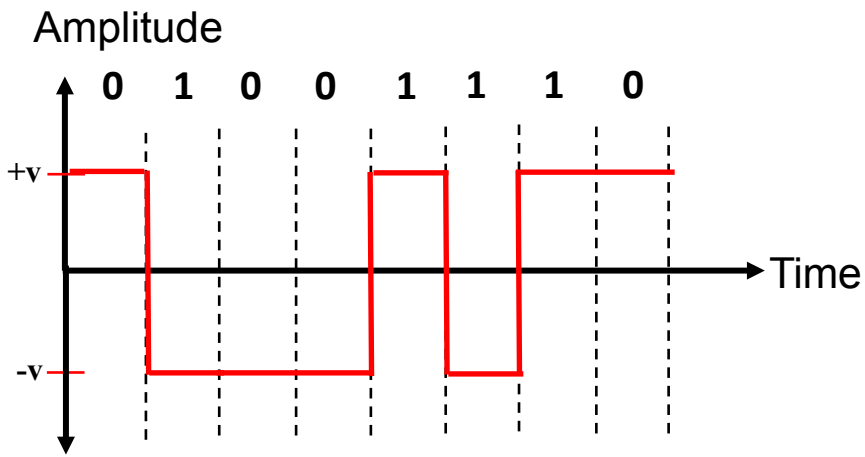


Polar NRZ - Level (NRZ-L)

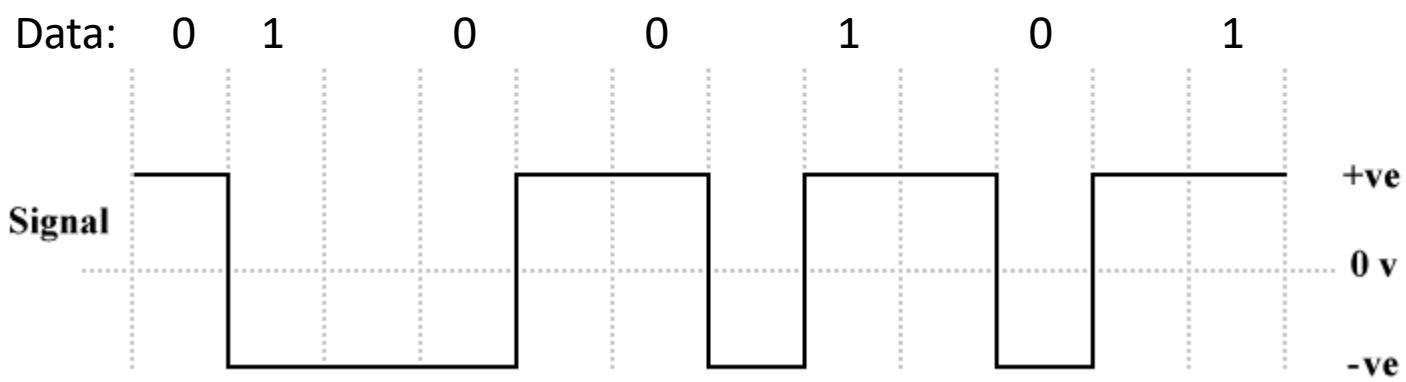


Polar NRZ - Inversion (NRZ-I)

Data: 01001110



Polar NRZ - Inversion (NRZ-I)



Polar RZ

Polar RZ coding works similarly to unipolar RZ but with a positive voltage for binary 1 and a negative voltage for binary 0 It returns to zero for the remaining half of the bit period. This approach helps with synchronization but requires more bandwidth.

- Advantages
  - Improved synchronization compared to NRZ
  - Better error detection
- Disadvantages
  - Higher bandwidth requirements

Polar RZ

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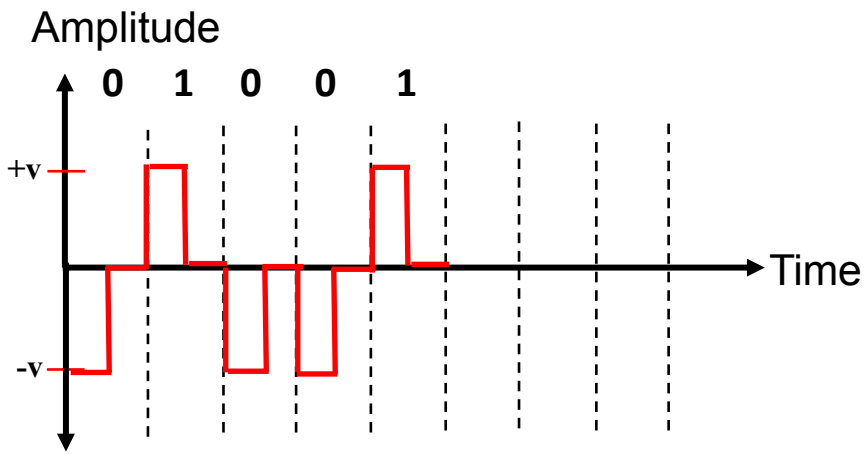
- Advantages
  - Improved synchronization compared to NRZ
  - Better error detection
- Disadvantages
  - Higher bandwidth requirements

Polar RZ

- The Return to Zero (RZ) scheme uses three voltage values. +, 0, -.
- Each symbol has a transition in the middle. Either from high to zero or from low to zero.
- This scheme has more signal transitions (two per symbol) and therefore requires a wider This scheme has more signal transitions (two per symbol) and therefore requires a wider bandwidth.
- No DC components or baseline wandering.
- Self synchronization - transition indicates symbol value.
- More complex as it uses three voltage level. It has no error detection capability

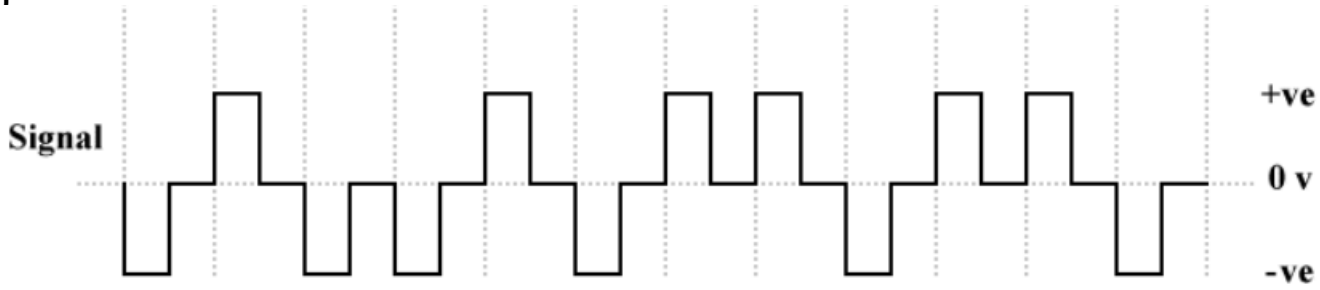
Polar RZ

Data: 01001



Polar RZ

Data: 0 1 0 0 1 0 1 1 0 1 1 0



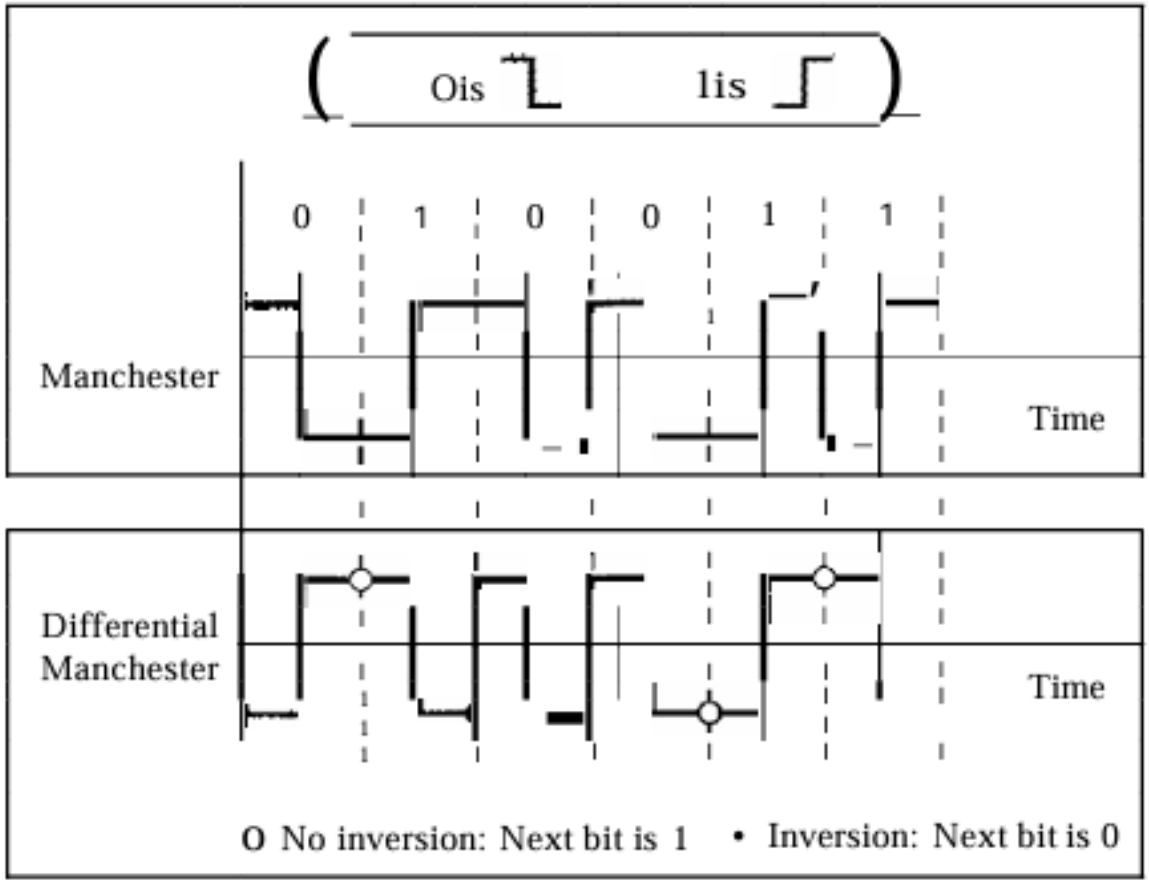
An example of polar RZ line coding

Biphase: Manchester and Differential Manchester

- Manchester coding consists of combining the NRZ-L and RZ schemes.
  - Every symbol has a level transition in the middle: from high to low or low to high. Uses only two voltage levels.

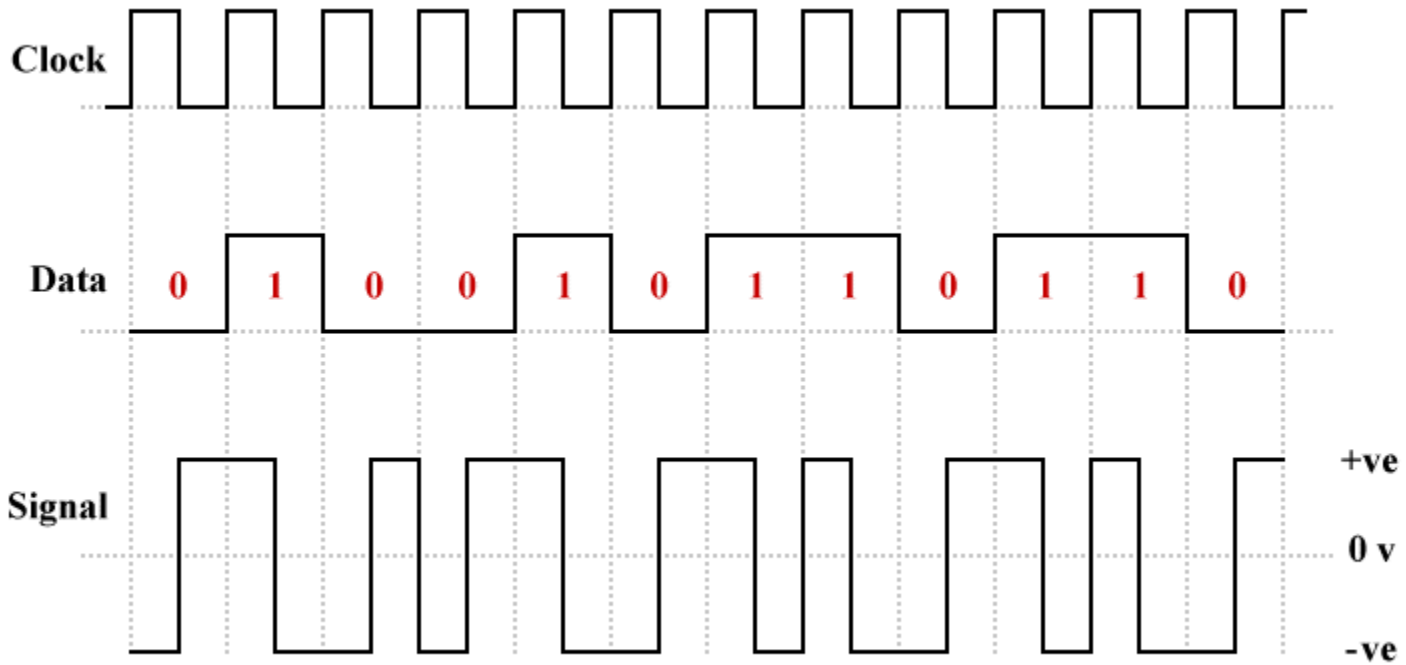
- Differential Manchester coding consists of combining the NRZ-I and RZ schemes.
  - Every symbol has a level transition in the middle. But the level at the beginning of the symbol is determined by the symbol value. One symbol causes a level change the other does not.

Biphase: Manchester and Differential Manchester



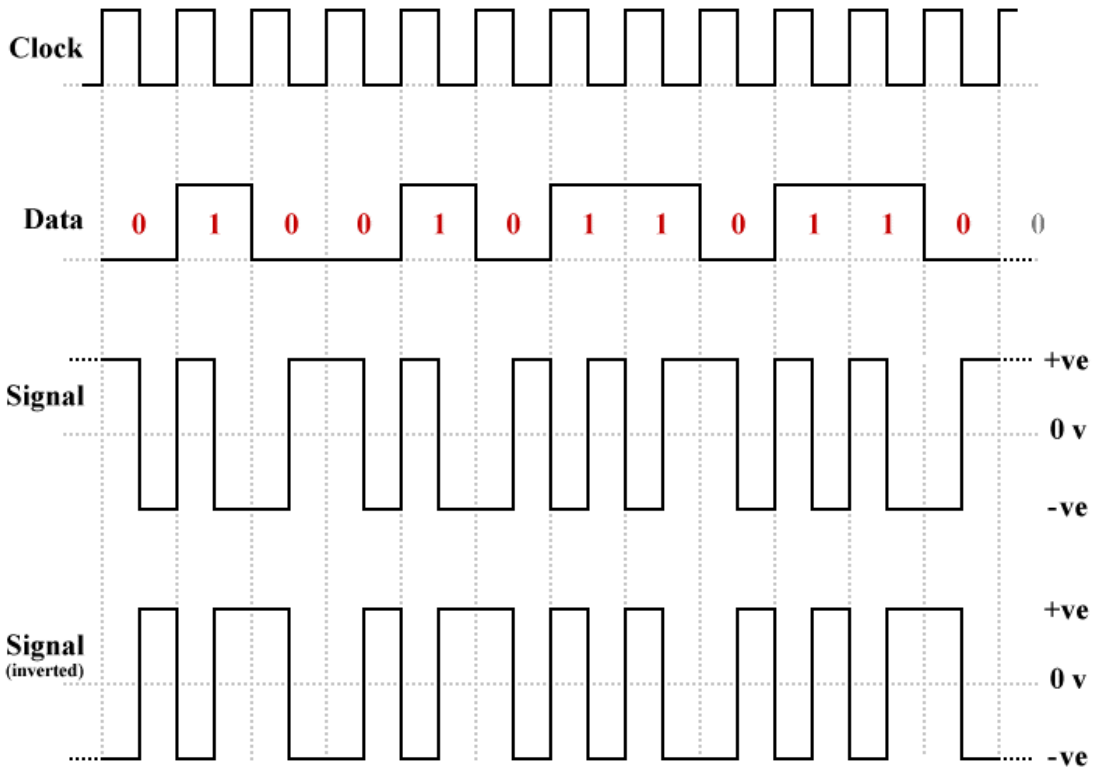
Biphase: Manchester

Data: 010010110110



Biphase: Differential Manchester

Data: 010010110110



Bipolar

Bipolar signaling uses three voltage levels positive zero and negative voltages. The binary 1 is represented by alternating between a positive and a negative voltage while binary 0 is represented by zero voltage

Advantages

- No DC component reducing signal degradation
- Better error detection and synchronization

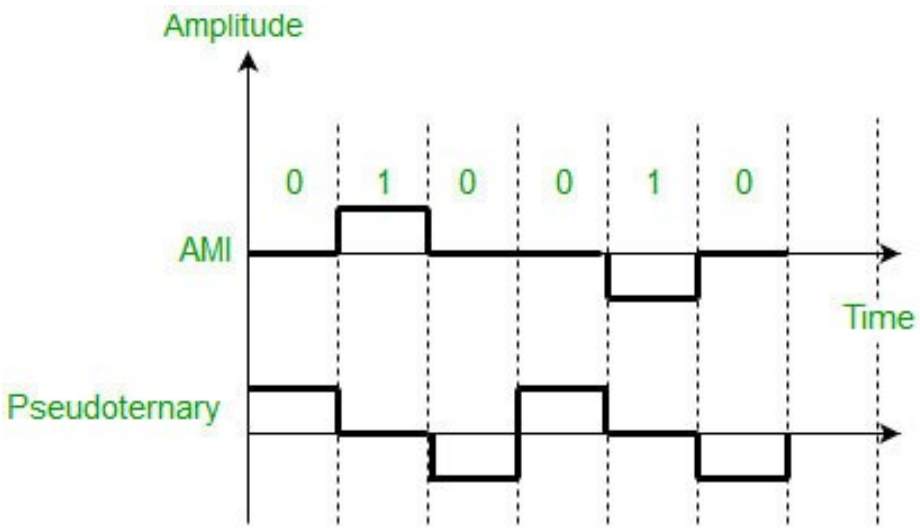
Disadvantages

- More complex to implement
- Requires additional processing for signal interpretation

Bipolar

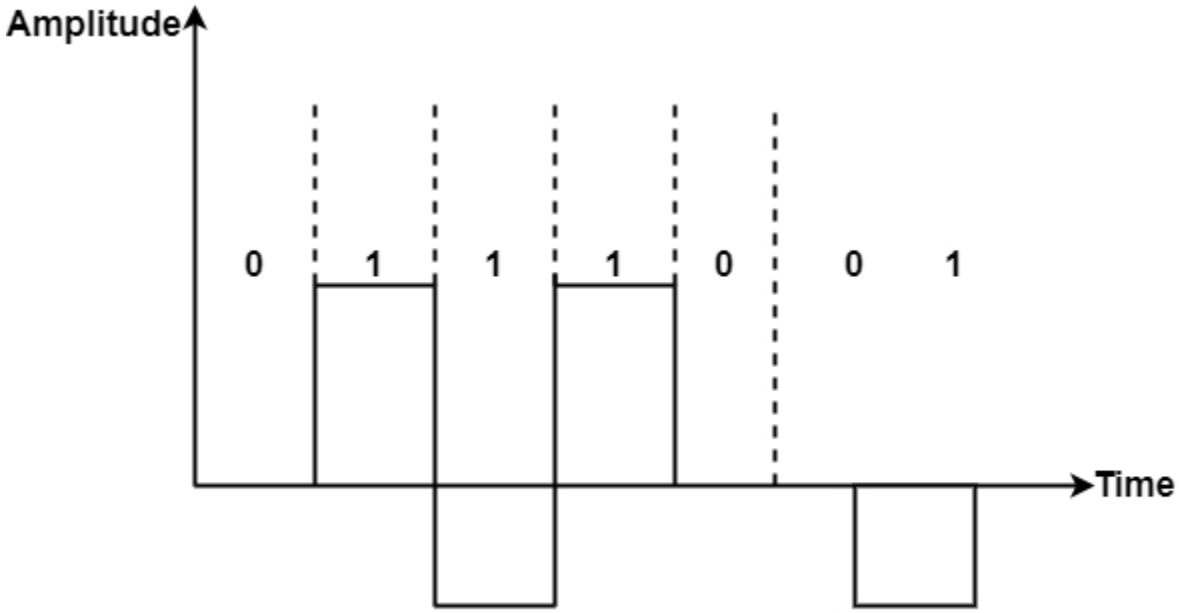
- **Alternate Mark Inversion (AMI) -** A neutral zero voltage represents binary 0. Binary 1's are represented by alternating positive and negative voltages.
- **Pseudoternary -** Bit 1 is encoded as a zero voltage and the bit 0 is encoded as alternating positive and negative voltages i.e., opposite of AMI scheme.

Example: Data= 010010



Alternate Mark Inversion (AMI)

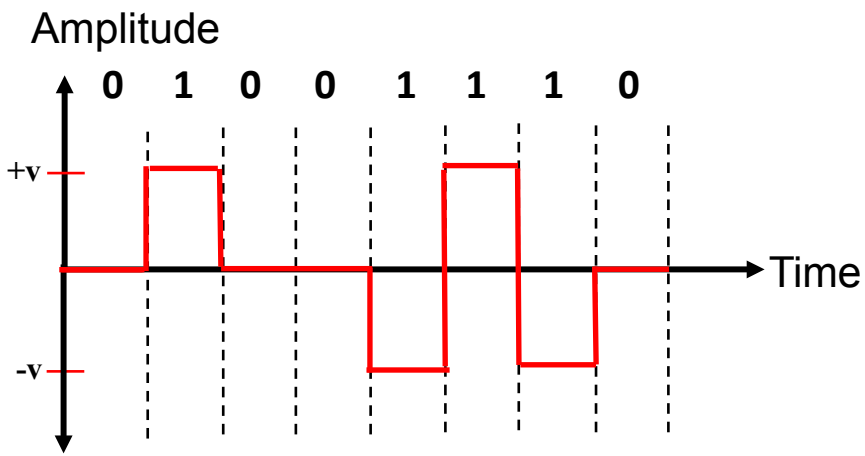
Data = 0111001



Bipolar Alternate mark inversion

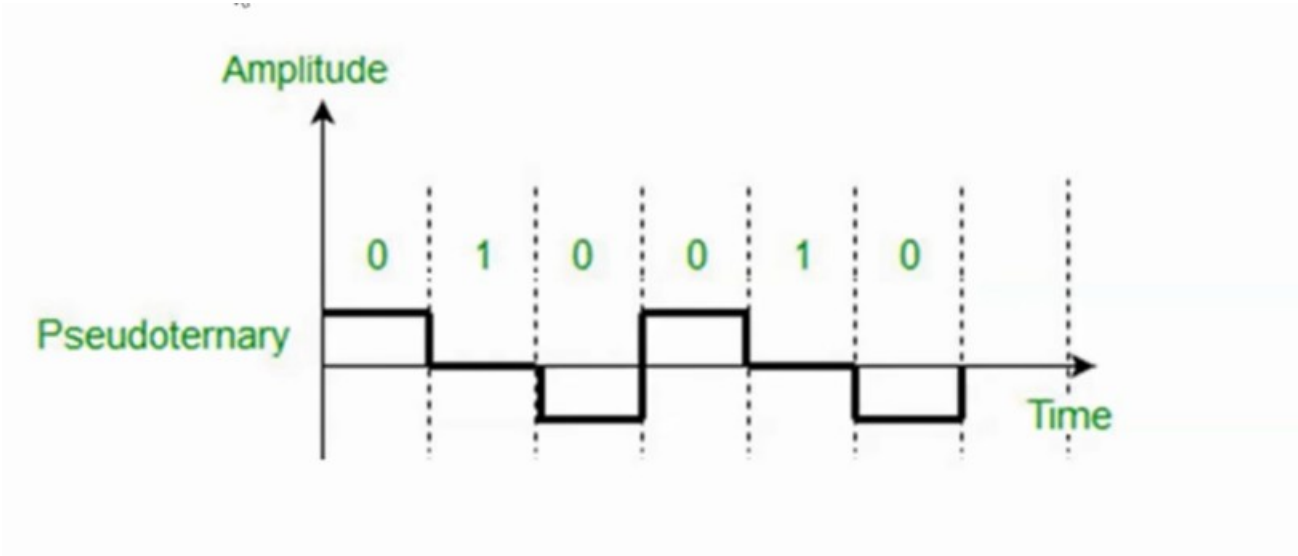
Alternate Mark Inversion (AMI)

Data = 01001110



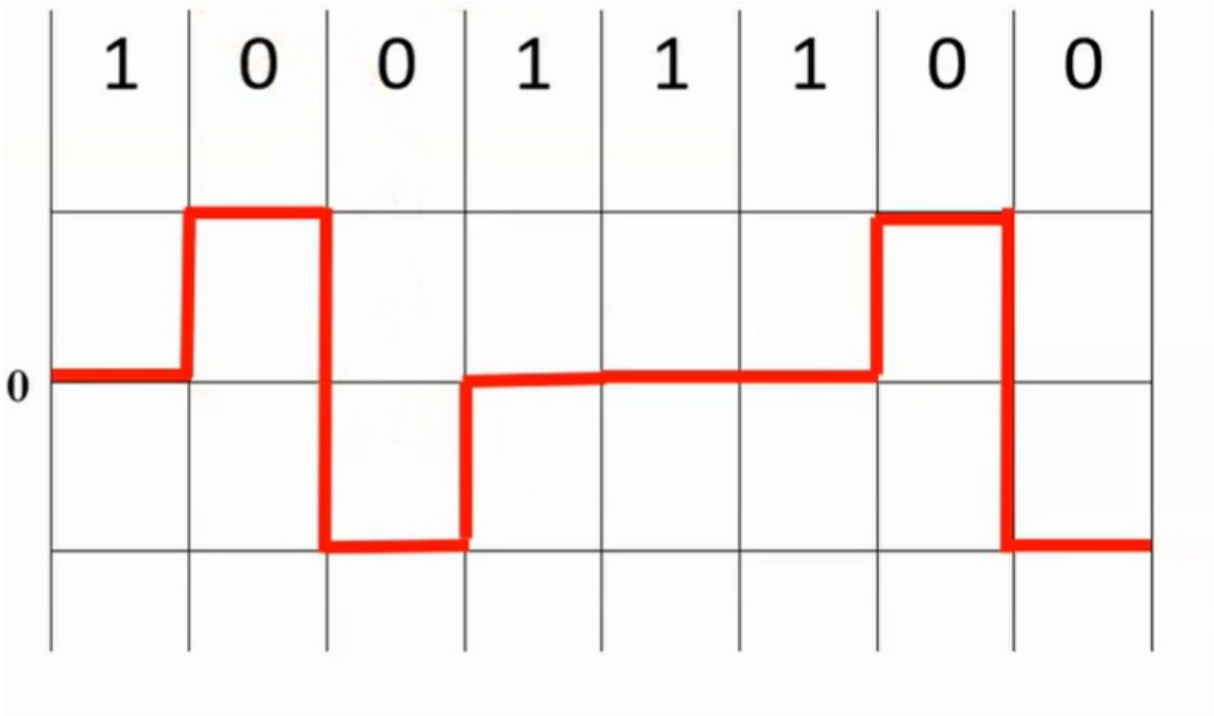
Pseudoternary

Data = 0111001



Pseudoternary

Data = 10011100



Analog Data Transmission

Converting digital data to a bandpass analog signal.is traditionally called digital to-analog conversion. Converting a low-pass analog signal to a bandpass analog signal is traditionally called analog-to-analog conversion.

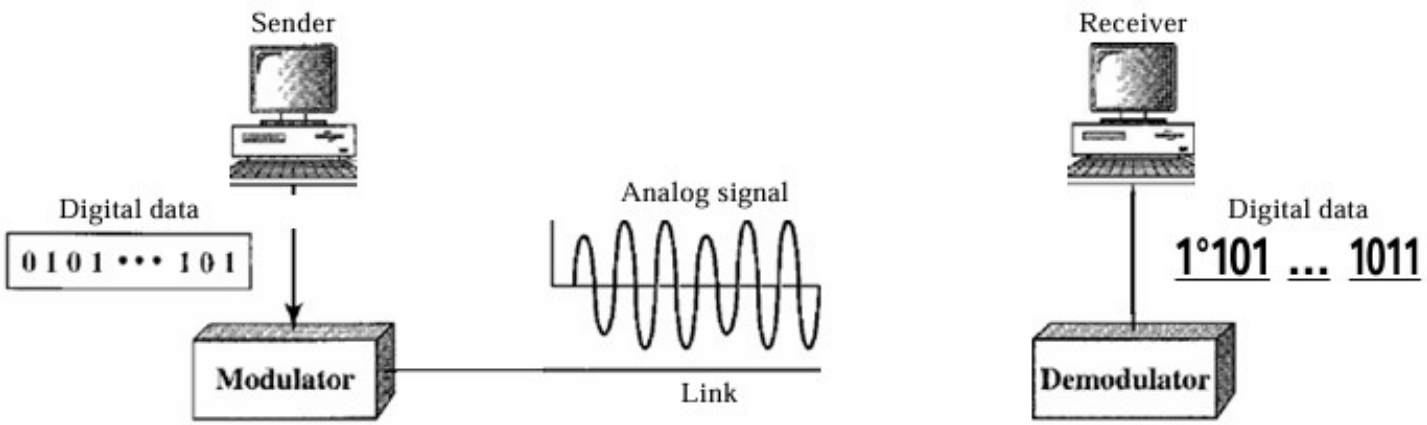


Figure Connecting devices

DIGITAL-TO-ANALOG CONVERSION

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure shows the relationship between the digital information, the digital-to-analog modulating process, and the resultant analog signal.

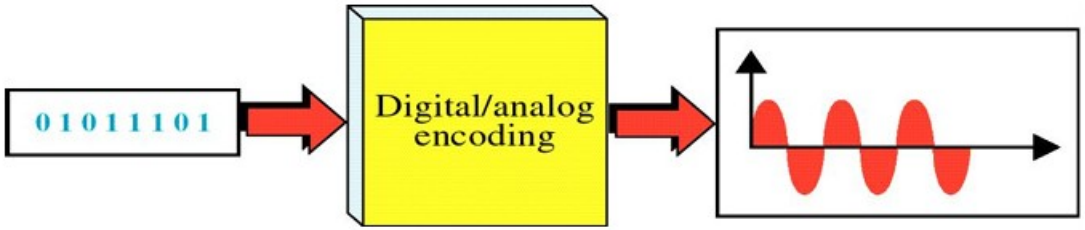
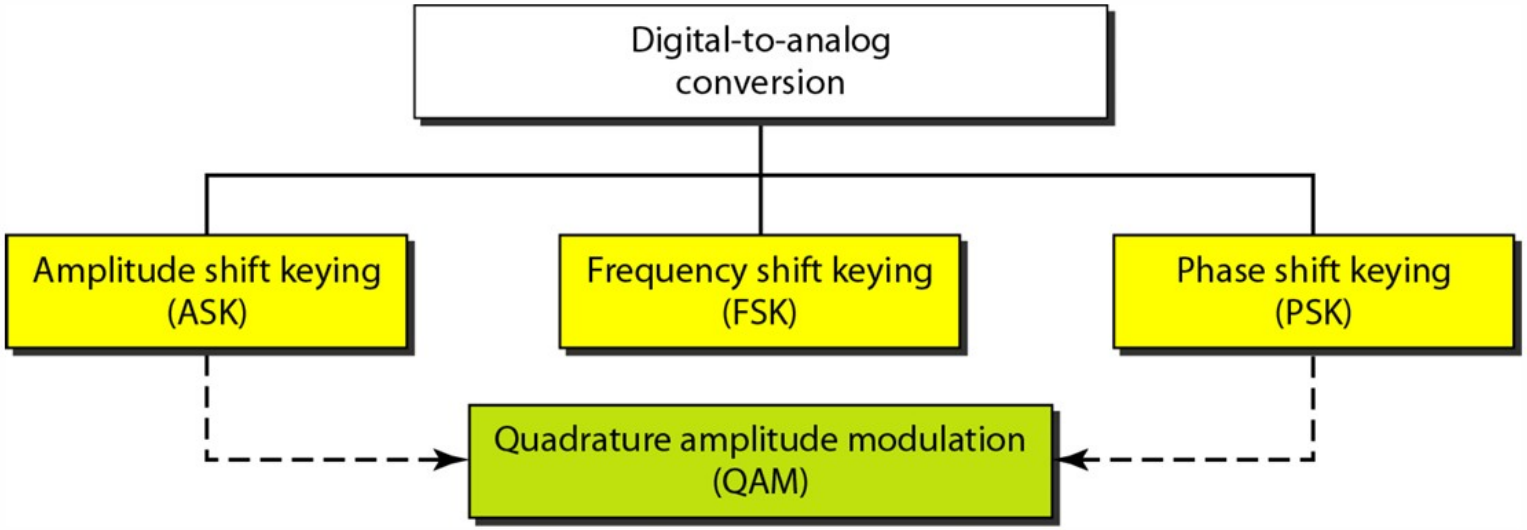


Figure Digital to Analog Modulation

DIGITAL-TO-ANALOG CONVERSION



Types of digital-to-analog conversion

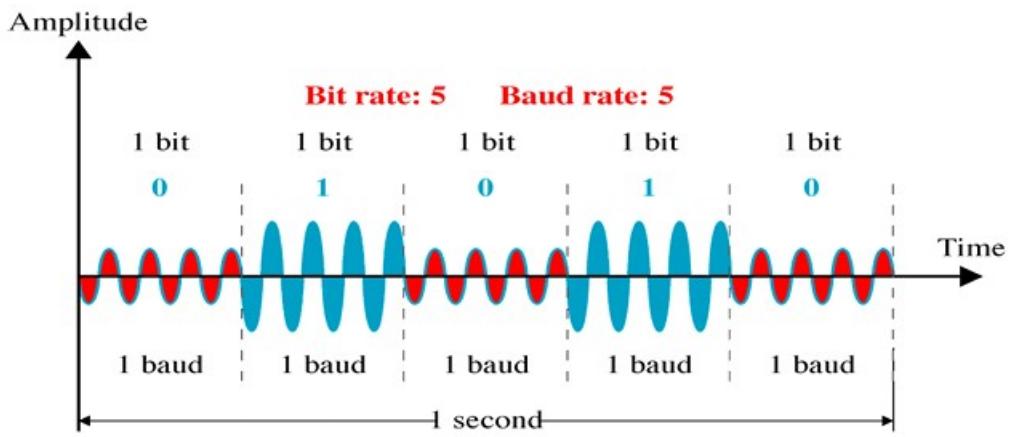
Digital-to-Analog Modulation Techniques

How digital data (Os and 1s) is sent over an analog channel (e.g., phone lines).

- ASK (Amplitude Shift Keying): Amplitude changes between high and low to represent 1 or 0.
- FSK (Frequency Shift Keying): Frequency changes to represent 1 or 0.
- PSK (Phase Shift Keying): Phase changes to represent 1 or 0.
- QAM (Quadrature Amplitude Modulation): Combines ASK and PSK for higher data rates.

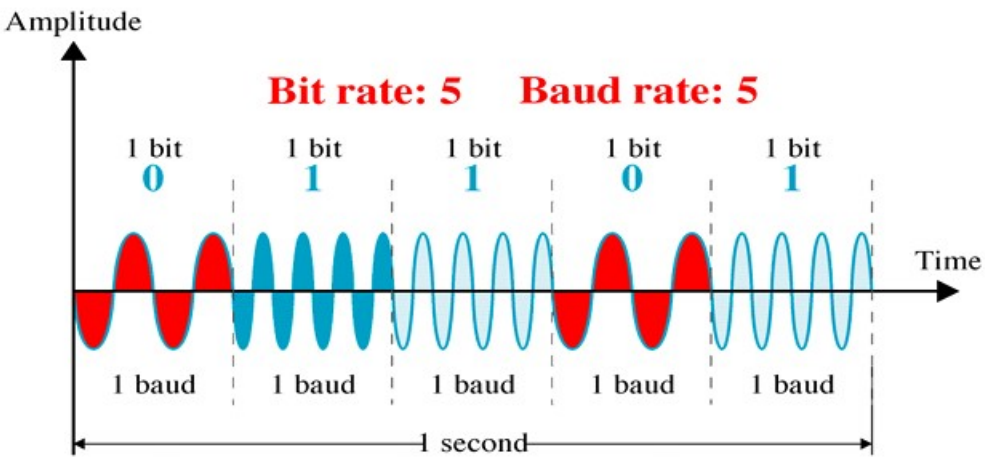
Amplitude Shift Keying (ASK)

ASK is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.
For example: a digital "1" could not affect the signal, whereas a digital "0" would, by making it zero.
The line encoding will determine the values of the analog waveform to reflect the digital data being carried.



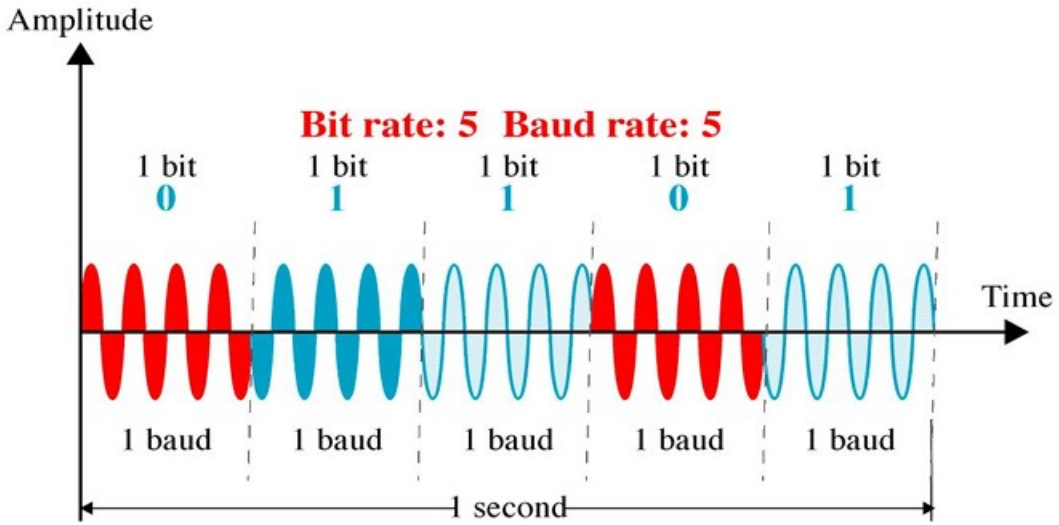
Frequency Shift Keying (FSK)

- In FSK the frequency of the carrier signal is varied to represent the binary 1 or 0. Both
- Amplitude and phase remain constant, while the frequency changes. FSK avoids most of
- the noise problems of PSK.



Phase Shift Keying (PSK)

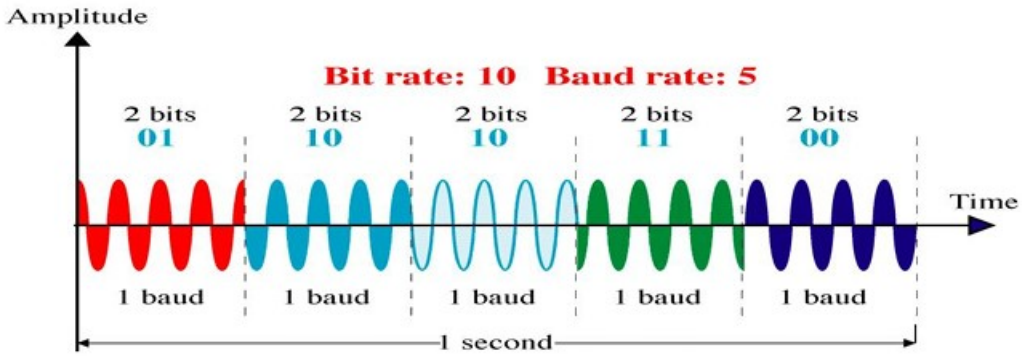
- In PSK the phase of the carrier is varied to represent binary 1 or 0.
- Both peak amplitude and frequency remains constant as the phase changes.



Quadrature Amplitude Modulation (QAM)

- To increase the bit rate, we can code 2 or more bits onto one signal element.
- In QPSK, we parallelize the bit stream so that every two incoming bits are split up and PSK a carrier frequency. One carrier frequency is phase shifted 90° from the other - in quadrature.
- The two PSKed signals are then added to produce one of 4 signal elements. L = 4 here.
- This Technique is also called Q-PSK .
- The pair of bits represented by each phase is called a di bit.

The constellation diagram for the signal in shown in following figures:



ANALOG-TO-ANALOG CONVERSION

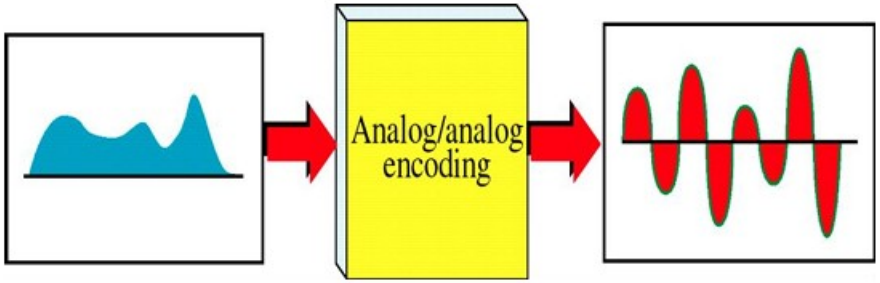


Figure: Analog to Analog Modulation

ANALOG-TO-ANALOG CONVERSION

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure shows the relationship between the digital information, the digital-to-analog modulating process, and the resultant analog signal.

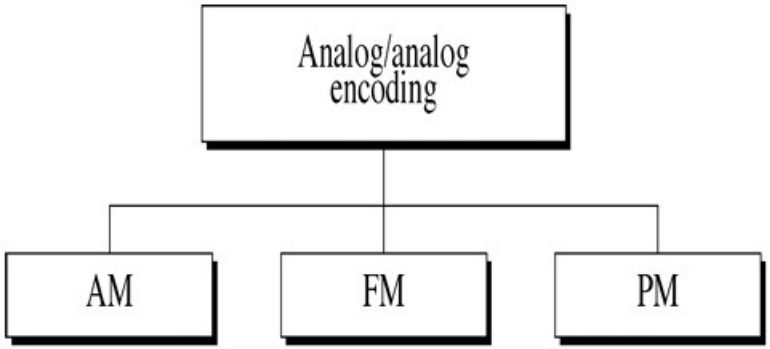
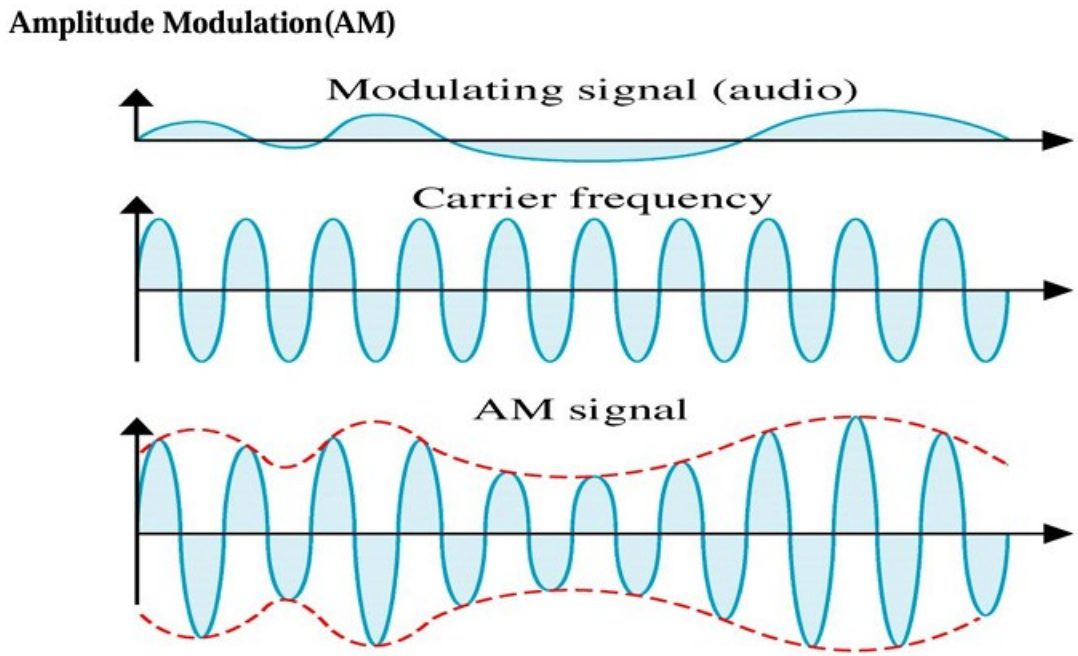
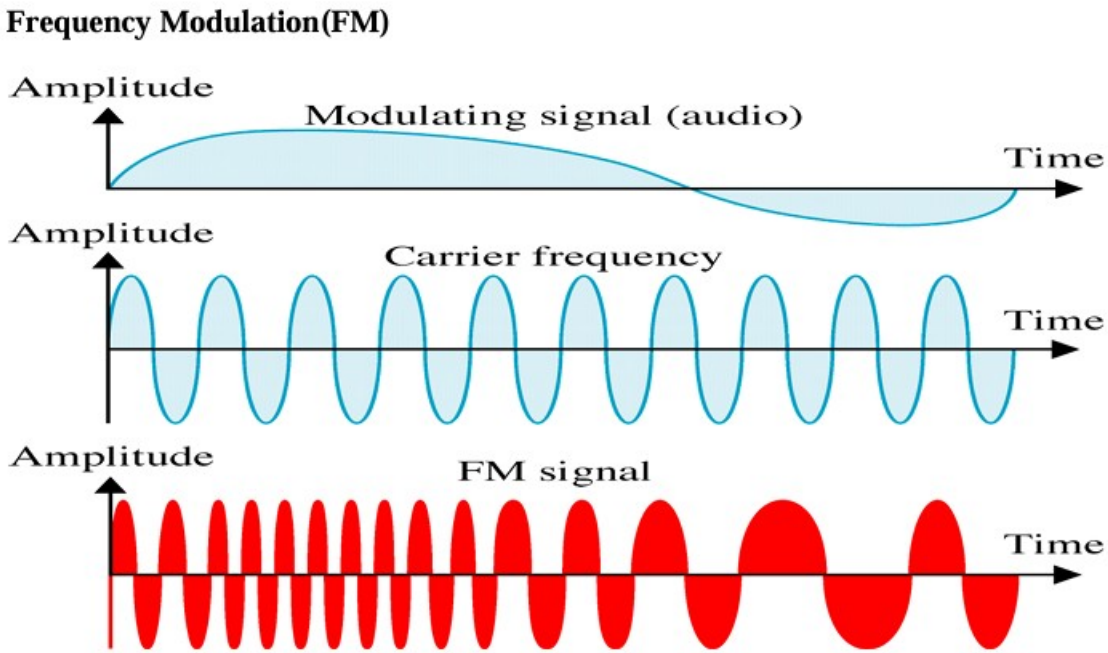


Figure Digital to Analog Modulation

Amplitude Modulation(AM)



Frequency Modulation(FM)



Frequency Modulation(FM)

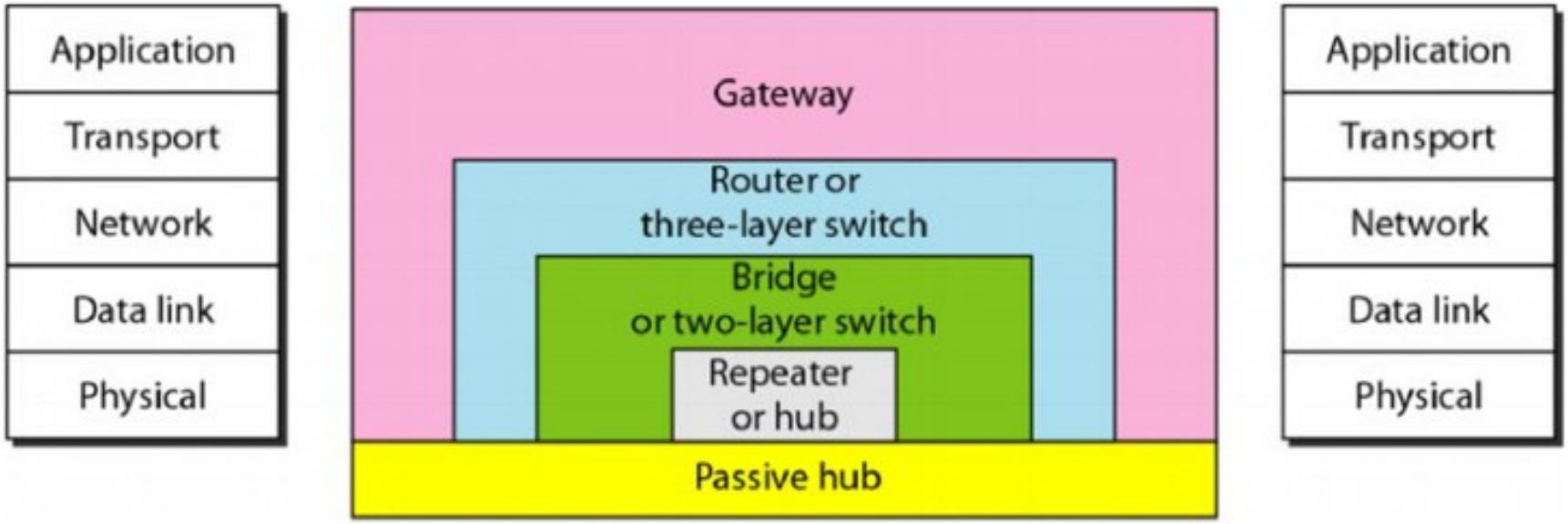
Due to simpler hardware requirements, phase modulation(PM) is used in some systems as an alternative to frequency modulation.

CONNECTING DEVICES

- Repeaters
- Hubs
- Switches
- Bridges
- Router
- Gateways
- Network Interface Cards (NICs)
- Wireless access points
- Modems

CONNECTING DEVICES

Connecting devices are divided into 5 different categories based on the layer in which they operate in the network.



Five categories of connecting devices

Fig. 12 Connecting devices

HUB

- A hub is used as a central point of connection among media segments.
- A Hub is a hardware device used to connect several computers together.
- A hub contains multiple ports. Cables from network devices plug in to the ports on the hub.

Types of HUBS :–

- A passive hub is just a connector. It connects the wires coming from different branches.
  - The signal pass through a passive hub without regeneration or amplification.
  - Connect several networking cables together
- Active hubs or Multiport repeaters- They regenerate or amplify the signal before they are retransmitted.

HUB

- Hubs Acts on the physical layer Operate on bits rather than frames also called multiport repeater
- Used to connect stations adapters in a physical star topology but logically bus Connection to the hub consists of two pairs of twisted pair wire one for transmission and the other for receiving.
- Hub receives a bit from an adapter and sends it to all the other adapters without implementing any access method. does not do filtering (forward a frame into a specific destination or drop it) just it copy the received frame onto all other links.

HUB

- The entire hub forms a single collision domain, and a single Broadcast domain
  - Collision domain: is that part of the network (set of NICs) when two or more nodes transmit at the same time collision will happen.
  - Broadcast domain: is that part of the network (set of NIC) where each NIC can 'see' other NICs' traffic broadcast messages.
- Multiple Hubs can be used to extend the network length
- For 10BaseT and 100BaseT the maximum length of the connection between an adapter and the hub is 100 meters.

HUB

- **Hub Advantages:**
  - Simple, inexpensive device
  - Multi-tier provides graceful degradation
  - Portions of the LAN continue to operate if one hub malfunctions
  - As an active hubs regenerate signals, it increases the distance that can be spanned by the LAN (up to 100 meters per segment).
  - Hubs can also be connected locally to a maximum of two other hubs, thereby increasing the number of devices that can be attached to the LAN.
  - Active hubs are usually used against attenuation, which is a decrease in the strength of the signal over distance.

HUB

- **Hub Disadvantages:**
  - Bandwidth is shared by all hosts i.e. 10Mbs shared by 25 ports/users.
  - Can create bottlenecks when used with switches.
  - Have no layer 3 switching capability.
  - Most Hubs are unable to utilize VLANS
  - Individual segment collision domains become one large collision domain (reduce the performance) can't interconnect different Ethernet technologies (like 10BaseT & 100BaseT) because no buffering at the hub

REPEATER

- A physical layer device the acts on bits not on frames or packets
- Can have two or more interfaces
- When a bit (0,1) arrives, the repeater receives it and regenerates it, the transmits it onto all other interfaces.

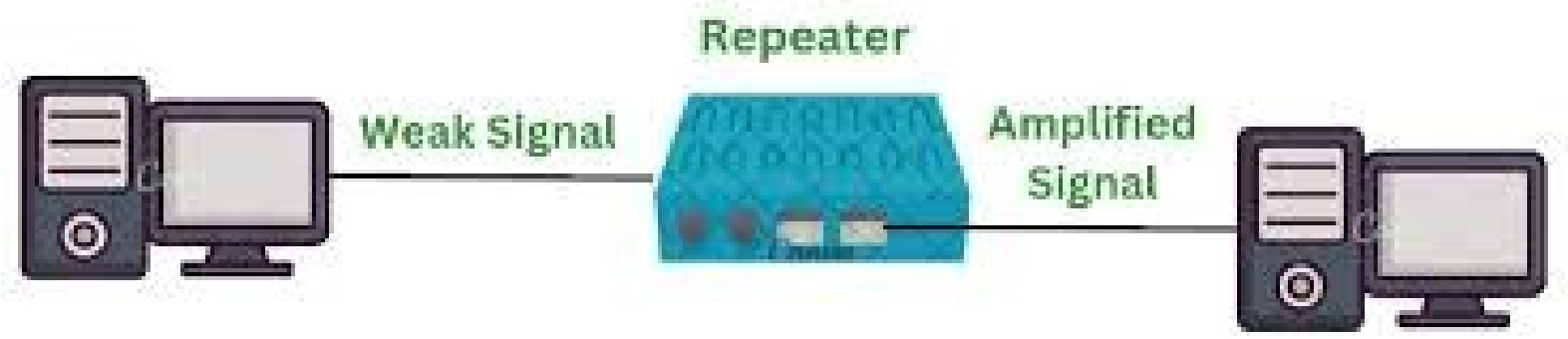


Fig. (a) Repeater

REPEATER

- Used in LAN to connect cable segments and extend the maximum cable length- extending the geographical LAN range
  - Ethernet 10base5 – Max. segment length 500m – 4 repeaters (5 segments) are used to extend the cable to 2500m)
  - Ethernet 10Base2- Max. segment length 185m - 4 repeaters (5 segments) are used to extend the cable to 925m
- Repeaters do not implement any access method
  - If any two nodes on any two connected segments transmit at the same time collision will happen.

happy

LEARNING